COMP 333 — End-to-End Data Analytics Pipeline

Executive Summary

Ronnie Chan (27206003) · Patrice Gallant (40301020) · Nesrine Larbi (40079009) | Concordia University | April 2026

1. Project Overview

This project implements a complete, end-to-end data analytics and machine learning pipeline applied to **New York City (NYC) Yellow Taxi Trip Records** for June–July 2025. **Phase 1** covers data acquisition, wrangling, EDA, and baseline modelling; **Phase 2** advances into feature engineering, supervised classification, unsupervised clustering, and result interpretation.

Attribute	Detail
Primary Dataset	NYC Yellow Taxi Trip Records — June & July 2025 (~7.3 M rows, 1.4+ GB on disk)
Supplementary	NYC Hourly Weather (Open-Meteo API) — temperature (°C) and precipitation (mm)
Environment	Google Colab (12.7 GB RAM); Python 3 · pandas · scikit-learn · XGBoost

2. Phase 1 — Data Acquisition, Cleaning & Baseline Modelling

Data Wrangling & EDA Highlights

- Removed 4 exact duplicates and **150,444 mirrored records**; dropped **238,440 zero-distance rows**; imputed missing values and applied Winsorization (99th-percentile) to five financial features.
- Final clean dataset: **14 features** across ~7 M records (Parquet). Trip distance is the strongest fare predictor (r = 0.87); credit card accounts for ~75 % of rides; demand peaks post-17:00 on weekdays.
- **Baseline (Phase 1):** Linear Regression Model 2 (*trip_distance* + *tip_amount*) selected for low MSE, high R^2, and stable generalisation.

3. Research Questions

RQ1 — Supervised Learning	RQ2 — Unsupervised Learning
"Can we classify a high-tipper passenger based on trip features (distance, time of day, day of week, rate code, passenger count) and weather conditions?" Target: **is_high_tip** (tip ≥ 20 % of fare_amount) — Models: **Random Forest, XGBoost**	*"Can we identify distinct natural clusters of NYC taxi trips based on temporal and financial features to study customer behaviour?"* Method: **PCA** + **MiniBatch K-Means** (k = 7)

4. Phase 2 — Feature Engineering

Thirty features were engineered across three groups: **(1)** log-transforms and cyclical hour/day encodings; **(2)** domain flags (is_rush_hour, is_airport, is_credit_card, fare_per_mile); **(3)** polynomial/interaction terms (distance², distance × passengers, temp × precipitation). Filter, embedded (RF importance), and wrapper (RFE) methods produced a **top-12 composite feature set (FINAL_FEATURES)**, excluding leakage variables (*tip_amount*, *total_amount*, *tip_ratio*).

5. Supervised Learning Results (RQ1)

Both models trained on **1 M stratified credit-card records** (70/15/15 split, RandomSearchCV 3-fold, F1 metric). **XGBoost selected** for marginally higher AUC and 40 % faster training.

Metric	Random Forest	XGBoost
CV / Val F1-Score	0.8637 / 0.8636	0.8637 / 0.8636
Test Accuracy	0.77	0.77
Recall — High Tip	1.00	1.00

AUC-ROC	0.6293	0.6302
Training Time	~250 s	~150 s

- Top features: **improvement_surcharge**, **fare_amount**, **congestion_surcharge** — tipping is driven by payment method and fare structure, *not* time of day or weather.
- Limitation: persistent 75/25 class imbalance causes low recall (0.08) for 'Low Tip'; cash rides introduce structural label noise. Future: SMOTE or undersampling.

6. Unsupervised Learning Results (RQ2)

PCA reduced the 8-feature space to **6 PCs (≈ 90 % variance)**; elbow method on a 100 K subsample identified **k = 7**. Full dataset (~7 M rows) clustered with **MiniBatch K-Means** (Silhouette = 0.18; Davies-Bouldin = 1.35 — overlapping clusters consistent with real-world behavioural data).

Cluster	Label	Key Characteristic
0	Long Trip	Highest mean trip distance — airport or inter-borough transfers
1, 2	Short Trip	Low distance; likely artificial split of one natural group
3	Medium Trip	Mid-range distances; no strongly differentiating secondary feature
4	Weather-Impacted	Very high mean precipitation — rainy-day demand spike
5	Peak Morning	Lowest mean pickup hour — morning commute segment
6	High-Occupancy	Highest mean passenger count — group trips

7. Conclusions & Actionable Insights

- **RQ1 — Tipping Behaviour:** XGBoost classifies high-tippers with 77 % accuracy (AUC 0.63). Tipping is driven by **payment method and fare economics** — not weather or time of day. Action: Integrate the model as a real-time scoring layer in dispatch applications to flag high-tip-probability trips (credit-card, airport-rate, high-fare), enabling drivers to make informed route-acceptance decisions and fleets to prioritise these rides.
- **RQ2 — Trip Archetypes (Pricing & Dispatch):** Seven clusters expose distinct demand segments. Actions: (a) Apply **dynamic surge pricing** during the Weather-Impacted cluster (Cluster 4) to balance supply and willingness-to-pay during rain events; (b) Pre-position vehicles in business districts before 08:00 to capture the Peak-Morning commute segment (Cluster 5); (c) Offer **group-ride promotions** to High-Occupancy trips (Cluster 6) to grow average revenue per vehicle; (d) Investigate whether Clusters 1 and 2 are a genuine split or an artefact — merging them could sharpen short-trip targeting.
- **Addressing Class Imbalance (Future Work):** The persistent 75/25 split suppresses recall for Low-Tip rides (0.08). Action: Apply **SMOTE oversampling** or cost-sensitive weighting in the next iteration to improve minority-class detection without sacrificing overall accuracy.
- **Pipeline Scalability:** The end-to-end pipeline processed 1.4+ GB within Colab's 12.7 GB RAM through dtype downcasting, GC discipline, and stratified sampling. Action: Migrate batch training to a cloud environment (e.g., GCP Vertex AI or AWS SageMaker) to remove the RAM bottleneck and enable monthly retraining on fresh TLC data.